Session 10 (AIML) - Assignment Questions

Q1. (Basic Class and Object)
Create a class named Car with:

- Attributes: brand and model (set using __init__).
- A method display_info() that prints the brand and model.

Create two objects of the Car class with different values and call the method for both.

Q2. (Using **init** Constructor)
Write a class Book that takes title, author, and price as parameters in __init__.
Create a method show_details() to print all the book information.
Create at least two book objects and display their details.

Q3. (Instance Methods and self)
Create a class BankAccount with:

- Attributes: account_holder and balance (initialized in __init__).
- Methods:
  - deposit(amount) — add amount to balance.
  - withdraw(amount) — subtract amount from balance (only if sufficient balance).
  - show_balance() — print current balance.

Create an object and demonstrate all methods.

Q4. (Method with Parameters)
Create a class Student with attributes name and marks.
Add a method calculate_grade() that returns "Pass" if marks ≥ 40, otherwise "Fail".
Create 3 student objects, display their details, and print their grades.

Q5. (Multiple Methods)
Create a class Employee with:

- Attributes: name, salary (use __init__).
- Methods:
    - display_details()
    - give_raise(amount) — increase salary and print new salary.
    - yearly_bonus() — return 10% of current salary.

Create an employee object and demonstrate all methods.

Q6. (Real-world Object Modeling)
Create a class MobilePhone with attributes: brand, model, battery_percentage.
Add these methods:

- charge(percent) — increase battery.
- use_phone(minutes) — reduce battery by 1% per 10 minutes.
- show_status() — print current status.

Create an object and simulate some usage.

Q7. (Combining Concepts)
Create a class Rectangle with attributes length and width.
Add methods:

- area() — return the area.
- perimeter() — return the perimeter.
- display() — print length, width, area, and perimeter.

Take input from user for length and width, create object, and display results.

Q8. (Updating Attributes)
Create a class Player with attributes name, score, and level.
Add methods:

- increase_score(points)
- level_up() — increase level by 1.

- show_progress()

Create an object, update score and level multiple times, and show progress.

Q9. (Debugging OOP Code)
The following code has errors. Correct it and add proper comments explaining the fixes:

Python

```
class Person:
    def __init__(name, age):
        name = name
        age = age

    def introduce():
        print("My name is" name "and I am" age "years old.")

p1 = Person("Rahul", 25)
p1.introduce()
```

Q10. (Mini Project - Combined OOP)
Create a **Simple Library Management System** using OOP:

Design a class Book with:

- Attributes: title, author, status ("Available" or "Issued").
- Methods: issue_book(), return_book(), show_info().

Create multiple book objects and demonstrate issuing/returning books with proper messages.
Use a menu (while loop) to allow the user to:

1. Add a new book
2. Issue a book
3. Return a book
4. Show all books

5. Exit